SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:25**

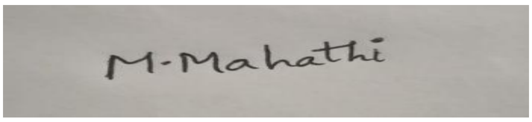
Concept Mapping
Annexure A

Criteria	Concept Identification ***(2.5)***	Linkages ***(2.5)***	Organization ***(2)***	Integration of Literature ***(2)***	Clarity ***(1)***	***Total(10)***
Weightage	*25%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I __M.Mahathi(192324098)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Concept Mapping					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Annexure A

Concept Map 1: Decision Support Systems (DSS)

Task Description:

Construct a concept map that explains how a Decision Support System (DSS) supports organizational decision-making.

🎯 What your map must show:

1. Core components: Data, Model, User Interface
2. Types of DSS
3. Flow: Data → Analysis → Decision
4. Role in business decision-making
5. Real-world example linkage

Concept Map 2: Operational Systems vs DSS

Task Description:

Develop a comparative concept map that clearly distinguishes Operational Systems (OLTP) and Decision Support Systems (OLAP).

What your map must show:

1. Definitions of OLTP and DSS
2. Differences (data type, purpose, users, processing)
3. Similarities and interactions
4. Real-time vs historical data
5. Example systems (banking vs analytics)

Concept Map 3: Architecture of Data Warehouse

Task Description:

Create a hierarchical concept map illustrating the architecture of a Data Warehouse.

What your map must show:

1. Data Sources → ETL → Data Warehouse → Data Marts

2. Metadata layer
3. Front-end tools (reporting, OLAP)
4. Data flow direction
5. Role of each layer

Concept Map 4: OLAP, Data Cubes & Dimensional Modeling

Task Description:

Design a concept map that connects OLAP operations, Data Cubes, and Dimensional Modeling techniques.

🎯 What your map must show:

1. Data cube structure (dimensions, measures)
2. OLAP operations (slice, dice, roll-up, drill-down)
3. Star schema and snowflake schema
4. Fact and dimension tables
5. Analytical usage

Concept Map 5: Query Processing & BI Tools (KNIME)

Task Description:

Construct a concept map explaining how query processing integrates with BI tools, focusing on KNIME for reporting and OLAP applications.

What your map must show:

1. Query processing stages (Parsing, Optimization, Execution)
2. Role of BI tools
3. KNIME workflow structure
4. Data → Query → Visualization pipeline
5. OLAP application integratio

UNIT I

INTRODUCTION TO DATAWAREHOUSING

1 DECISION SUPPORT SYSTEM (DSS)

- Purpose: Support decision making using data and models
- Features: User oriented, ad-hoc query, analytical, non-volatile
- Components: Data, Model, Knowledge, User Interface
- Benefits: Better decisions, improved efficiency, strategic advantage

2 OPERATIONAL VS DECISION SUPPORT SYSTEM

Aspect	OLTP	DSS
Purpose	Day-to-day ops	Decision making
Data	Current, detailed	Historical, integrated
Users	Clerks, staff	Managers, execs
Processing	Simple, transactional	Complex, ad-hoc
Volume	GBs	TBs/PBs
Design	Normalization	Denormalization

3 ARCHITECTURE OF DATA WAREHOUSE

- Flow: Sources → ETL → Data Warehouse → OLAP Server → Front-End Tools
- Sources: OLTP DBs, Files, APIs, External | Tools: Reports, Dashboards, Analytics
- Characteristics: Subject Oriented, Integrated, Time Variant, Non-volatile

4 ETL

EXTRACT: Collect data from multiple sources

TRANSFORM: Cleanse, validate, conform, enrich

LOAD: Store data in data warehouse

5 OLAP AND DATA CUBES

- OLAP Operations:
- Drill-Down
- Roll-Up
- Slice
- Dice
- Pivot

- Data Cube dims:
- Product
- Location
- Time

- Benefits:
- Multidimensional view
- Fast complex analysis
- Better insights

6 DIMENSIONAL DATA MODELING

Star Schema

Denormalized — central fact table with surrounding dimension tables

Dim Dim

Fact

Dim Dim

Snowflake Schema

Normalized — dimensions further split into related sub-tables

Sub Sub

Fact

Sub Sub

7 QUERY PROCESSING

- 1. Parse Query
- 2. Optimize Query
- 3. Execute Query
- 4. Generate Results

8 OPEN-SOURCE BI TOOLS

- Tools: Pentaho, KNIME, Metabase, Apache Superset, Jaspersoft
- Used for: Data Integration, Visualization, Reporting, Dashboards, Analytics

9 OLAP APP USING KNIME

- Visual analytics platform; workflows with no coding
- Data integration, transformation, analysis & reporting
- Connectors for DBs/files; supports OLAP ops & visualizations

10 DATA CUBE COMPUTATION

- MOLAP: multidimensional cubes — very fast performance
- HOLAP: cubes + relational detail (hybrid)
- ROLAP: relational tables — scalable, flexible
- Star Cube (Star Schema) | Snowflake Cube (Snowflake Schema)